

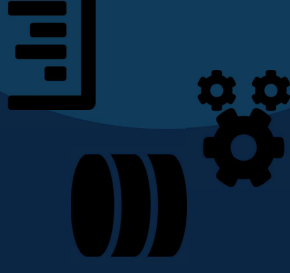
DATA SCIENCE AND ANALYTICS

Data Analysis Pipeline

UNIT I

From Raw Data to Knowledge Discovery

3 HOURS



Topics Overview

1.1

Knowledge Discovery from Database (KDD)

Understanding the KDD process and its stages

1.2

Structured & Unstructured Data

Differentiating data types with real-world examples

1.3

Overview of Data Preprocessing

1.3.1 Data Cleaning | 1.3.2 Data Integration | 1.3.3 Data Transformation & Discretization



1.1

Knowledge Discovery from Database

(KDD)

Definition

Knowledge Discovery in Databases (KDD) is the non-trivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data. It was introduced by Fayyad, Piatetsky-Shapiro, and Smyth in 1996.

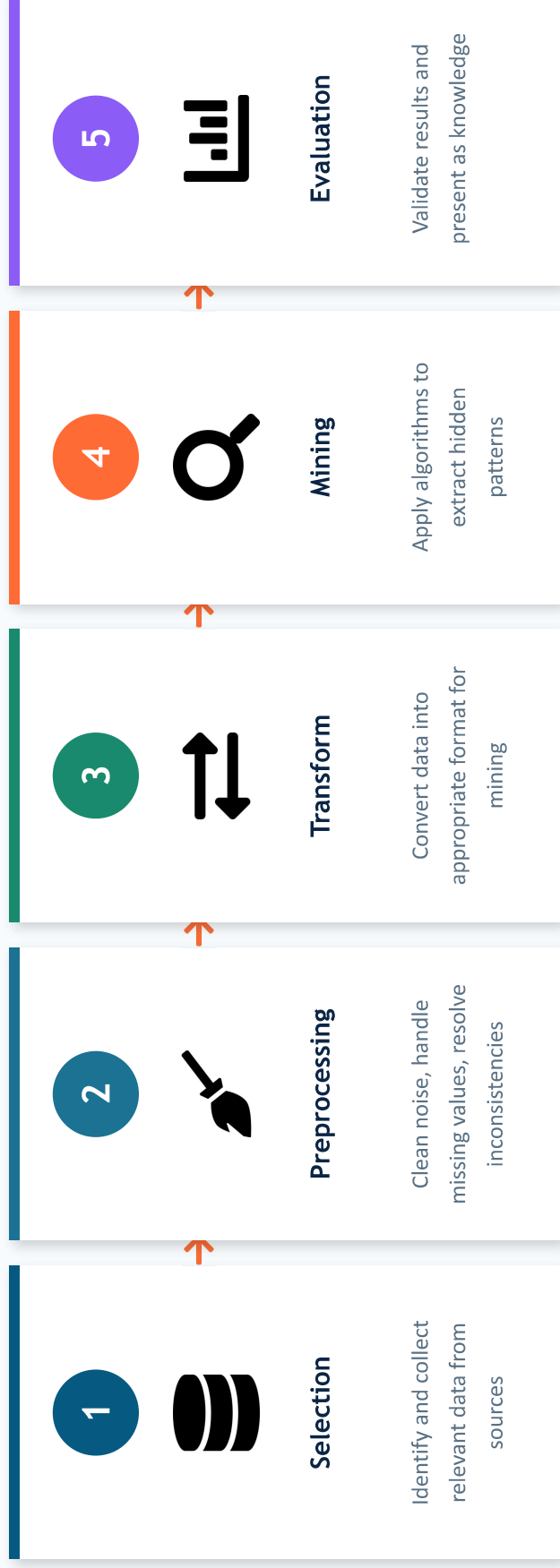
Key Characteristics

- Iterative and interactive process
- Multi-step methodology
- Data mining is a core step within KDD
- Goal: extract actionable knowledge

KDD vs Data Mining: Data Mining refers specifically to the pattern extraction step. KDD encompasses the entire process from data selection to interpretation of results.

The KDD Process

A systematic approach to discovering knowledge from data



KDD Steps in Detail

1. Data Selection

Identifying and selecting the target dataset from various data sources (databases, files, data warehouses). Focuses on relevant attributes and subsets.

2. Data Preprocessing

Handling missing values, removing noise and outliers, resolving inconsistencies. This step ensures data quality before transformation.

3. Data Transformation

Converting data into forms suitable for mining — normalization, aggregation, feature construction, dimensionality reduction.

4. Data Mining

Applying algorithms (classification, clustering, association, regression) to discover patterns and relationships in the transformed data.

5. Evaluation & Interpretation

Assessing discovered patterns using interestingness measures, visualizing results, and consolidating validated knowledge for decision making.



1.2

Structured & Unstructured Data

Understanding data forms

Types of Data

Data exists in various formats that affect how it can be stored, processed, and analyzed. Understanding these formats is crucial for choosing appropriate preprocessing and mining techniques.

Structured

Organized in fixed schema (rows & columns). Easy to query with SQL.

Relational DBs, Spreadsheets, CSV files

Semi-Structured

Has some organizational properties but no rigid schema.

JSON, XML, HTML, Email, NoSQL

Unstructured

No predefined format or organization. Most data falls here (~80%).

Text, Images, Audio, Video, Social media posts

Structured vs Unstructured Data

Criteria	Structured Data	Unstructured Data
Format	Predefined schema (tables)	No predefined format
Storage	RDBMS, Data Warehouses	NoSQL, Data Lakes, File Systems
Query	Easy (SQL)	Complex (NLP, ML required)
Examples	Financial records, Inventory	Emails, Videos, Social media
Volume	~20% of enterprise data	~80% of enterprise data
Analysis	Statistical, BI tools	AI/ML, Deep Learning
Scalability	Vertical scaling	Horizontal scaling

Real-World Data Examples

Structured Data

- Relational databases (MySQL, PostgreSQL)
- Excel spreadsheets and CSV files
- Financial transaction records
- Student enrollment records
- Census and survey data
- Inventory management systems

Unstructured Data

- Emails and text documents
- Social media posts (Facebook, Twitter/X)
- Images, audio, and video files
- Sensor data and IoT streams
- Medical imaging (X-rays, MRI)
- Web pages and blog content

1.3

Overview of Data Preprocessing

Preparing data for analysis through cleaning, integration, and transformation



1.3.1

Data Cleaning



1.3.2

Data Integration



1.3.3

Transformation &
Discretization

Why Is Data Preprocessing Important?

"Garbage In, Garbage Out" — the quality of analysis depends entirely on data quality.

60%
of data issues

Incomplete Data

Missing values, missing attributes, lacking certain attributes of interest

25%
of data issues

Noisy Data

Errors, outliers, and inconsistencies due to faulty instruments or human error

15%
of data issues

Inconsistent Data

Discrepancies in naming conventions, codes, or formats across sources

1.3.1 Data Cleaning

Data cleaning (or data cleansing) deals with detecting and correcting (or removing) corrupt, inaccurate, or irrelevant records from a dataset.

Missing Values

Ignore, fill with mean/median/mode, use global constant, or predict using regression/ML

Inconsistencies

Use domain knowledge, external references, or metadata to detect and fix discrepancies in naming/coding

Noisy Data

Binning (smoothing by bin means/boundaries), regression smoothing, clustering-based outlier detection

Duplicate Removal

Identify and merge duplicate records using record linkage and deduplication techniques

Handling Missing Values

Strategies for dealing with incomplete data

Method	Best When	Advantage	Limitation
Ignore the Tuple	Class label is missing	Simple to implement	May lose significant data
Fill Manually	Small datasets	Most accurate	Time-consuming, infeasible for large data
Global Constant	Quick placeholder needed	Easy to apply	May mislead mining algorithms
Attribute Mean/Median	Numerical attributes	Preserves distribution	Ignores relationships between attributes
Most Probable Value	Pattern exists in data	Best accuracy	Computationally expensive

Data Collection

Data collection is a systematic process essential for gathering information that can be analyzed to derive insights and support decision-making across various fields, including business, healthcare, and research.

Data collection involves acquiring observations or measurements systematically to address specific research questions or business needs.

It is the foundational step in data analysis, enabling organizations to make informed decisions based on accurate and relevant information

Types of Data Collection

Primary Data Collection: Involves gathering original data directly from sources.

This method is tailored to specific research objectives and includes techniques such as:

Surveys: Structured questionnaires distributed online, in person, or via phone to gather opinions or characteristics from a group.

Interviews: Direct interactions with respondents, which can be structured, semi-structured, or unstructured.

Experiments: Controlled studies to test hypotheses by manipulating variables and observing outcomes.



Types of Data Collection

Secondary Data Collection:

Utilizes existing data collected by others for different purposes. This method is often quicker and less expensive than primary data collection and includes sources such as:

Published Research: Academic journals, reports, and articles.

Online Databases: Repositories that provide access to statistical information and social surveys.

Government Records: Data maintained by governmental agencies that can be used for analysis

Data Collection Methods

Several methods are employed within these categories:

Surveys: Effective for understanding general characteristics or opinions.

Observations: Gathering data in natural settings without interference.

Web Scraping: Automated extraction of data from websites.

APIs: Interfaces that allow access to structured data from services.

Sensors: Devices that collect real-time data from the environment.

Steps in Data Collection

The process typically involves several key steps:

Define Research Objectives: Clearly articulate what information is needed.

Select Data Sources: Identify where relevant data can be obtained.

Choose Methods and Tools: Decide on the most suitable methods for collection.

Plan Procedures: Develop a detailed plan for how data will be collected and managed.

Collect Data: Implement the chosen methods while ensuring quality control

Data Issues

Data quality issues can significantly impact an organization's ability to make informed decisions and achieve operational efficiency.

Common Data Quality Issues

Inaccurate Data:

Description: Data that contains errors or does not reflect the true values.

Implications: Leads to poor decision-making and unreliable analytics.

Solutions: Implement robust data quality monitoring tools to identify inaccuracies and compare against known accurate datasets for correction



Data Issues

Incomplete Data:

Description: Records missing critical information (e.g., missing ZIP codes or demographic details).

Implications: Results in flawed analyses and operational challenges.

Solutions: Use data validation techniques during data entry and implement processes to fill in missing information

Data Issues

Duplicate Data:

Description: Multiple entries of the same record, which can inflate counts and skew analysis.

Implications: Wastes storage space and leads to confusion in reporting.

Solutions: Employ deduplication technologies to identify and merge or eliminate duplicate records

Data Issues

Inconsistent Formatting:

Description: Variations in how data is formatted (e.g., date formats).

Implications: Causes difficulties in data integration and analysis.

Solutions: Standardize formats across datasets and use automated tools to ensure consistency

Data Issues

Ambiguous Data:

Description: Data that can be interpreted in multiple ways due to unclear definitions or formatting.

Implications: Leads to misinterpretation and inaccurate reporting.

Solutions: Continuously monitor for ambiguity using autogenerated rules to clarify data meanings



Data Issues

Stale Data:

Description: Outdated information that no longer reflects current conditions.

Implications: Can result in missed opportunities or incorrect analyses.

Solutions: Regularly review and update datasets to ensure relevance

Data Issues

Unstructured Data:

Description: Data that lacks a predefined format, making it difficult to analyze (e.g., free-text fields).

Implications: Hinders effective data processing and analysis.

Solutions: Utilize natural language processing (NLP) tools to structure unstructured data for better usability

Data Issues

Hidden or Dark Data:

Description: Data that is collected but not used or analyzed, often overlooked in databases.

Implications: Represents missed insights and opportunities for improvement.

Solutions: Use data cataloging tools to uncover hidden data and assess its potential value

Data Issues

Data Overload:

Description: An overwhelming volume of data that complicates analysis efforts.

Implications: Can lead to analysis paralysis, where important insights are lost in the noise.

Solutions: Implement filtering techniques to prioritize relevant data for analysis

Data Issues

Human Error:

Description: Mistakes made during data entry or management processes.

Implications: Introduces inaccuracies and inconsistencies into datasets.

Solutions: Provide training on data management best practices and automate data capture where possible

Methods to handle incomplete data

1. Deletion Methods

Listwise Deletion: Remove all data for any observation that has one or more missing values. This is straightforward but can lead to a significant loss of data if many observations are incomplete.

Pairwise Deletion: Only exclude missing values when they are needed for a specific analysis, allowing for the use of available data across different variables. This method maximizes data usage but may lead to inconsistencies in results.

Dropping Variables: If a variable has a high proportion of missing data, it may be more effective to remove it entirely from the dataset, especially if it does not contribute significantly to the analysis

Methods to handle incomplete data

2. Imputation Techniques

Mean/Median/Mode Imputation: Replace missing values with the mean, median, or mode of the available data. This is simple but can reduce variability in the dataset.

K-Nearest Neighbors (KNN) Imputation: Estimate missing values based on the values of similar observations. This method considers relationships between variables and can provide more accurate imputations.

Regression Imputation: Use regression models to predict and fill in missing values based on other available data.

Multiple Imputation: Generate multiple datasets with different imputed values for the missing data and analyze them separately before combining results. This method accounts for uncertainty associated with missing values

Noisy Data

Noisy data refers to unwanted variations or distortions in datasets that obscure the underlying patterns or signals.

Noisy data contains irrelevant, erroneous, or random information that can arise from various sources, including measurement errors, data entry mistakes, hardware malfunctions, or environmental factors.

This noise can lead to misleading insights and affect model performance.

Types of Noisy Data

Random Noise: This type of noise is extraneous information that does not correlate with the underlying data. It often occurs due to inaccuracies in measurements or recording methods.

Misclassified Data: Data points that are incorrectly labeled or categorized, often due to human error during data entry or processing.

Outliers: Extreme values that deviate significantly from other observations in the dataset can distort analysis and model training.

Irrelevant Features: Features that do not contribute to the predictive power of the model can introduce noise and complicate analysis.

Measurement Errors: Variations caused by inaccuracies in sensors or instruments used for data collection.



Techniques for handling noisy data

Data Cleaning:

Filtering: Remove unwanted data points or categories from the dataset. This includes filtering out outliers and irrelevant features.

Binning: Grouping data into bins to reduce variance and smooth out noise by replacing values with bin means or medians.

Techniques for handling noisy data

Outlier Detection and Treatment:

Use statistical methods (e.g., Z-scores, IQR) to identify outliers and decide whether to remove them or cap their values.

Techniques for handling noisy data

Feature Selection:

Identify and eliminate irrelevant features using techniques such as:

Filter Methods: Statistical tests to assess feature relevance.

Wrapper Methods: Adding/removing features based on model performance.

Embedded Methods: Regularization techniques that incorporate feature selection within model training.

Techniques for handling noisy data

Regression Techniques:

Use regression analysis to determine relationships between variables and filter out noise by modeling the expected output based on input features.

Machine Learning Algorithms:

Implement robust algorithms that can handle noisy data effectively, such as decision trees or ensemble methods (e.g., random forests) which reduce sensitivity to noise.

Autoencoders:

Utilize neural networks designed for unsupervised learning to detect anomalies in datasets by reconstructing input data while filtering out noise.



Handling inconsistent data

Inconsistent data can arise from various sources and can lead to inaccurate analyses and unreliable results.

Handling inconsistent data is a crucial aspect of data cleaning and preprocessing in data science.

Syntactic inconsistencies: Variations in data formats (e.g., date formats like DD/MM/YYYY vs. MM/DD/YYYY).

Semantic inconsistencies: Discrepancies in meaning or interpretation of data values (e.g., different spellings for the same category).

Structural inconsistencies: Issues that arise when merging datasets with different schemas or structures.



Step to handle inconsistency

Identify Inconsistencies:

Use data profiling techniques to explore the dataset and identify anomalies, missing values, duplicates, and format discrepancies.

Tools like `.info()`, `.describe()`, and `.value_counts()` in Pandas can help summarize the dataset and reveal inconsistencies 2.

Define Consistency Rules:

Establish clear rules for what constitutes consistent data. This includes specifying formats, naming conventions, and acceptable value ranges 1.

For example, ensure all dates are formatted as "YYYY-MM-DD" or that categorical variables use standardized labels.



Step to handle inconsistency

Data Cleaning Techniques:

Standardization: Normalize data formats across the dataset using functions like `.replace()` or `.map()` in Pandas to correct known inconsistencies.

Deduplication: Remove duplicate entries that may inflate sample size or skew analysis results using methods such as `drop_duplicates()` in Pandas.

Recoding: Use functions to recode categorical variables that may have inconsistent spellings or categories (e.g., converting "old_category" to "corrected_category")

Step to handle inconsistency

Data Validation Checks:

Implement validation checks to ensure that data adheres to defined consistency rules. This includes checking for valid date ranges, numeric limits, and categorical values 13.

Use automated scripts or data quality tools to perform these checks regularly.

Historical Data Analysis:

Analyze historical data to identify patterns of inconsistency. This can help uncover recurring issues and inform preventive measures

1.3.2 Data Integration

Data integration merges data from multiple heterogeneous sources into a coherent data store — a data warehouse or a unified view.

Schema Integration

Matching attributes from different schemas. E.g.,
'customer_id' vs 'cust_no' — entity identification problem.

Data Value Conflicts

Resolving differences in representation: units (kg vs lb),
scales (0-1 vs 0-100), or encoding (M/F vs Male/Female).

Redundancy Detection

Identifying duplicate or derivable attributes using
correlation analysis (chi-square test, covariance).

Semantic Heterogeneity

Same attribute names with different meanings across
sources, or different names for the same entity.

1.3.3 Data Transformation

Data transformation converts data into appropriate forms suitable for mining. It involves strategies that consolidate data into forms appropriate for the mining process.

1 Smoothing

Remove noise using binning, regression, or clustering techniques.

2 Aggregation

Summarize data — e.g., daily sales aggregated to monthly or yearly totals.

3 Generalization

Replace low-level data with higher-level concepts using concept hierarchies.
E.g., street → city → country.

4 Normalization

Scale attribute values to a smaller range (e.g., 0 to 1) using min-max, z-score, or decimal scaling.

5 Feature Construction

Create new attributes derived from existing ones to aid the mining process.

Normalization Techniques

Min-Max Normalization

$$v' = (v - \min) / (\max - \min) \times (\text{new_max} - \text{new_min}) + \text{new_min}$$

Maps values linearly to a new range [new_min, new_max], typically [0, 1]. | Example: Income \$12K-\$98K $\rightarrow$ Normalized: (73.6K-12K)/(98K-12K) = 0.716

Z-Score Normalization

$$v' = (v - \mu) / \sigma$$

Normalizes based on mean (μ) and standard deviation (σ). Useful when min/max are unknown. | Example: If $\mu=54K$, $\sigma=16K$: (73.6K-54K)/16K = 1.225

Decimal Scaling

$$v' = v / 10^j \quad (\text{where } j = \text{smallest integer such that } \max(|v'|) < 1)$$

Normalizes by moving the decimal point. Value of j depends on max absolute value. | Example: Max value = 986 $\rightarrow j=3$: 986/1000 = 0.986

Discretization

Converting continuous attributes into discrete intervals

Discretization reduces the number of values for a continuous attribute by dividing the range into intervals. Interval labels replace actual data values.

Equal-Width Binning

Divides range into N bins of equal width. Width = $(\text{max} - \text{min}) / N$. Simple but sensitive to outliers.

Entropy-Based (Top-Down)

Uses information entropy to recursively find the best split points. Supervised — uses class labels.

Equal-Frequency Binning

Each bin contains approximately the same number of data points. Better handles skewed distributions.

Concept Hierarchy

Replaces numeric ranges with meaningful labels. E.g., age: youth (0-25), middle-aged (26-55), senior (56+).

Unit I Summary

- ✓ **KDD Process:** A systematic 5-step methodology: Selection → Preprocessing → Transformation → Mining → Evaluation
- ✓ **Data Types:** Structured (schema-driven, ~20%) vs Unstructured (free-form, ~80%) with Semi-Structured in between
- ✓ **Data Cleaning:** Handles missing values, noise, outliers, and inconsistencies to ensure data quality
- ✓ **Data Integration:** Combines data from multiple sources, resolving schema conflicts and redundancy
- ✓ **Transformation:** Smoothing, aggregation, normalization (min-max, z-score), and feature construction
- ✓ **Discretization:** Converts continuous values to intervals using binning, entropy, or concept hierarchies

Thank You

DATA SCIENCE AND ANALYTICS | UNIT 1: DATA ANALYSIS PIPELINE

Questions & Discussion